



Bài 03: Giới thiệu

Nhập Môn Ngành Công Nghệ Sinh Học Số

14.09.2026

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Viện Nghiên cứu Ứng dụng Khoa học Sức khỏe và Lão hóa – Bệnh viện Thống Nhất



Trân trọng chào đón các Tân Sinh Viên

Ngành Công Nghệ Sinh Học 2026

Nội dung

1. Giới thiệu dự án **BỘ GEN NGƯỜI & GIẢI TRÌNH TỰ GEN THỂ HỆ MỚI (NGS)**
2. Ví dụ mối quan hệ của **BIẾN THỂ GEN** và **BỆNH DI TRUYỀN**
3. Quy trình **XÉT NGHIỆM** gen bằng phương pháp giải trình tự thể hệ mới (NGS)
4. Năm ví dụ về **XÉT NGHIỆM** gen cho **BỆNH DI TRUYỀN** và **UNG THƯ**
5. Xét nghiệm gen trong bệnh nhiễm
6. Ứng dụng AI trong phân tích dữ liệu hình ảnh y khoa

Xin gửi lời cảm ơn trân thành đến

Đại học Đồng Tháp - Khoa Công Nghệ và Kỹ Thuật

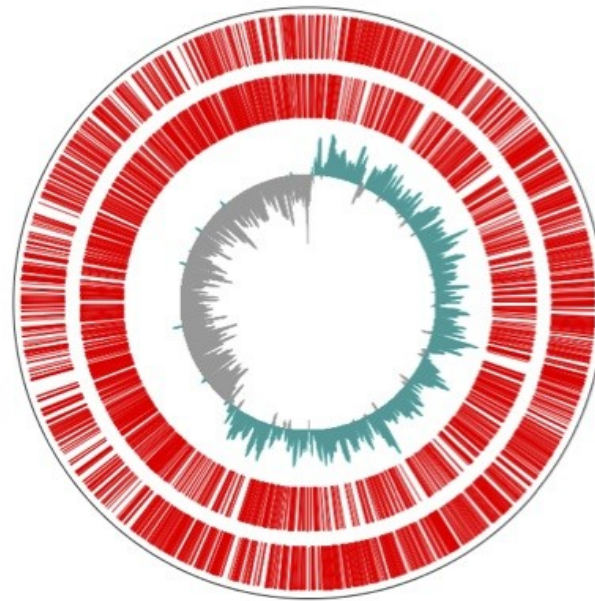
- **TS. Lê Uyển Thanh**
- **TS. Huỳnh Ngọc Tâm**
- **TS. Trần Đức Tường**

Ứng dụng NGS trong bệnh nhiễm

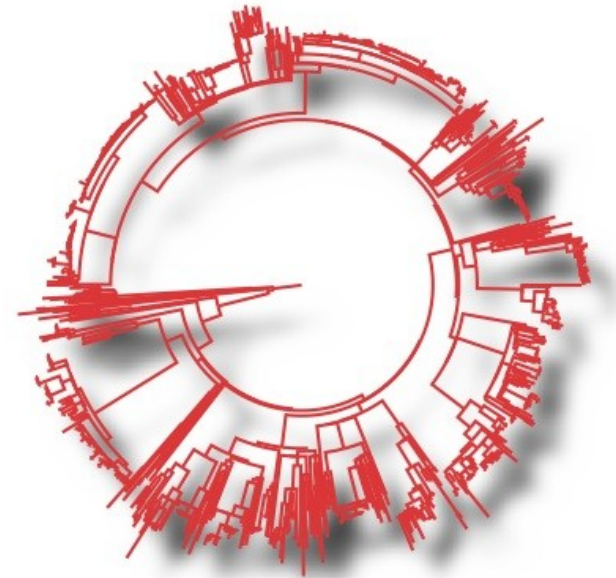
Phân tích bộ gen vi khuẩn: nghiên cứu một tác nhân gây bệnh ở mỗi mẫu phân lập



Phân lập



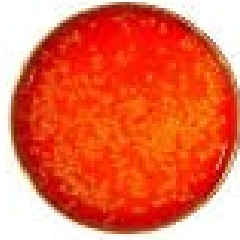
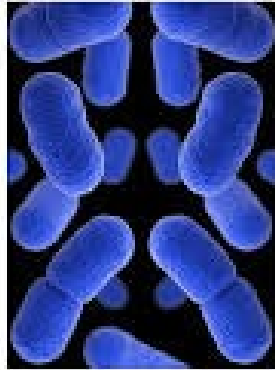
Giải trình tự bộ gen



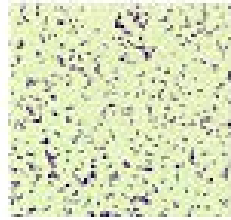
Serotyping/genotyping

Listeria monocytogenes

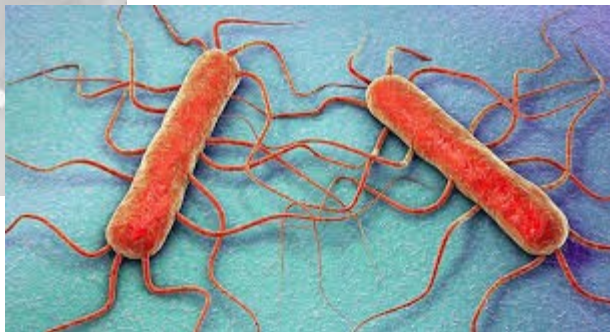
microbeonline



Beta-hemolytic colonies



Gram positive coccobacilli



Listeria monocytogenes EGD-e, complete genome

GenBank: AL591824.1

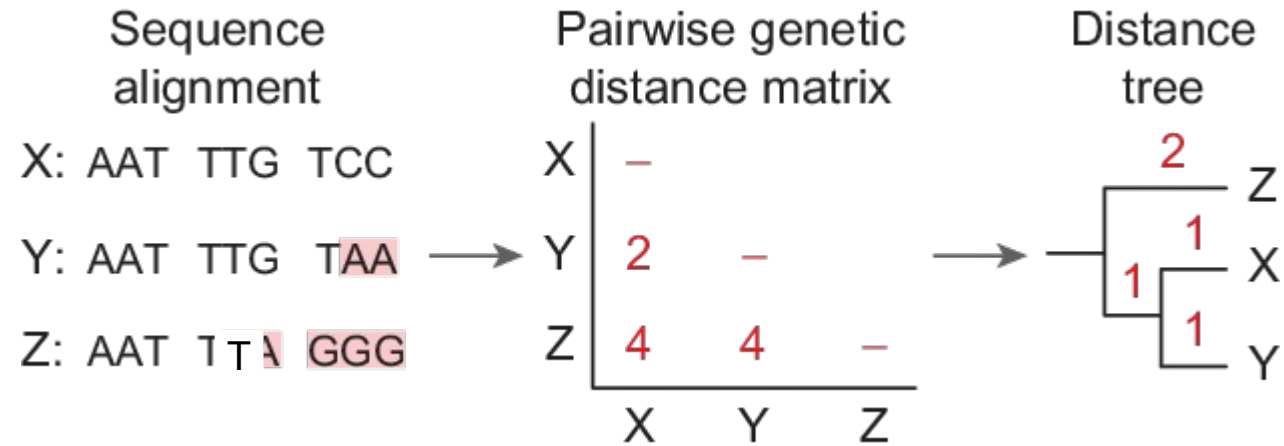
[GenBank](#) [Graphics](#)

>AL591824.1 *Listeria monocytogenes* EGD-e, complete genome

```
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```

<https://www.ncbi.nlm.nih.gov/nuccore/AL591824.1?report=fasta>

Building Phylogenetic Tree



Building Phylogenetic Tree

(a)

Sp 1: TTCATTGCCA

Sp 2: ATCATCAGGA

Sp 3: TTCGTCGCAA

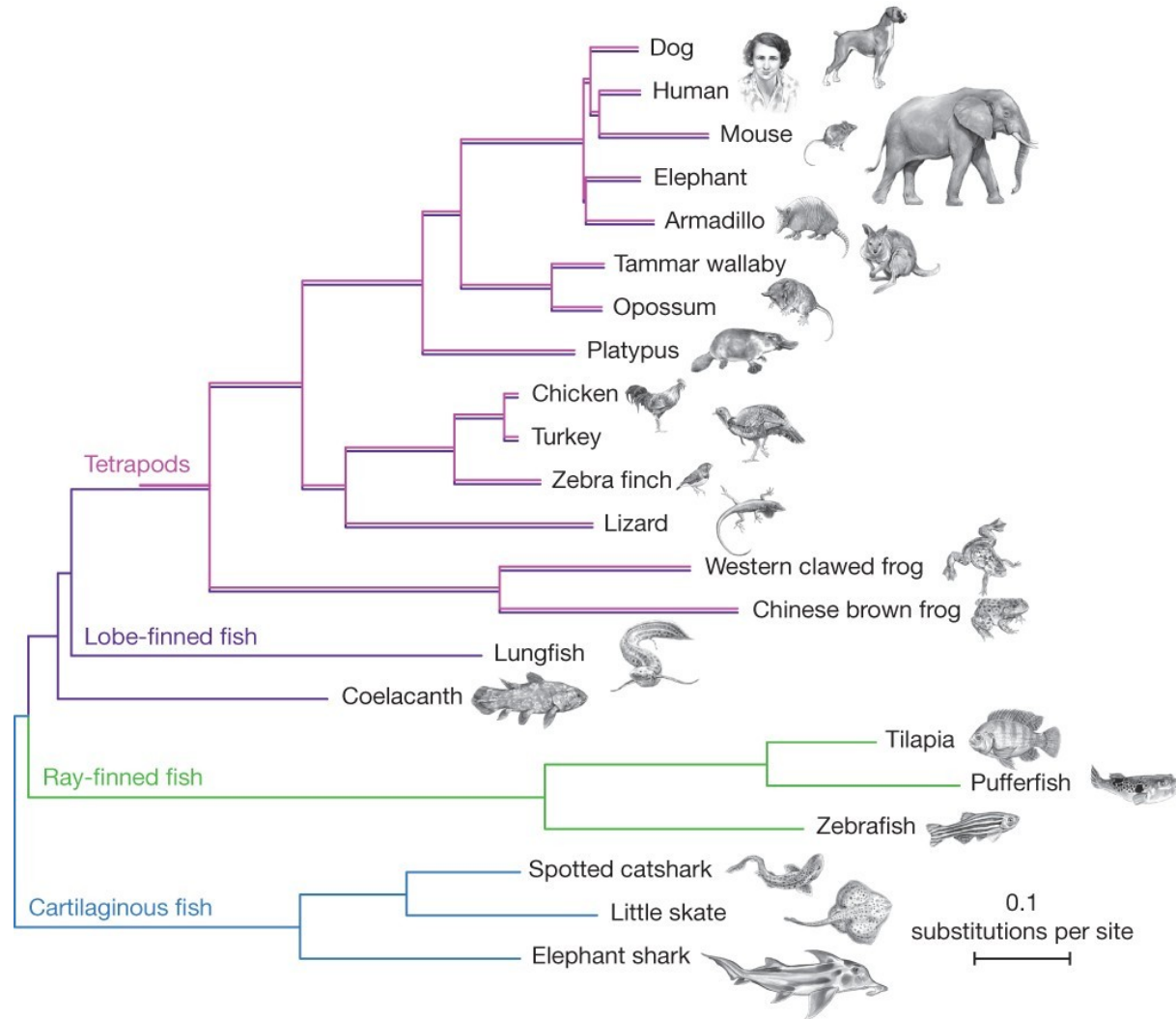
Sp 4: CTCGTCGAGG



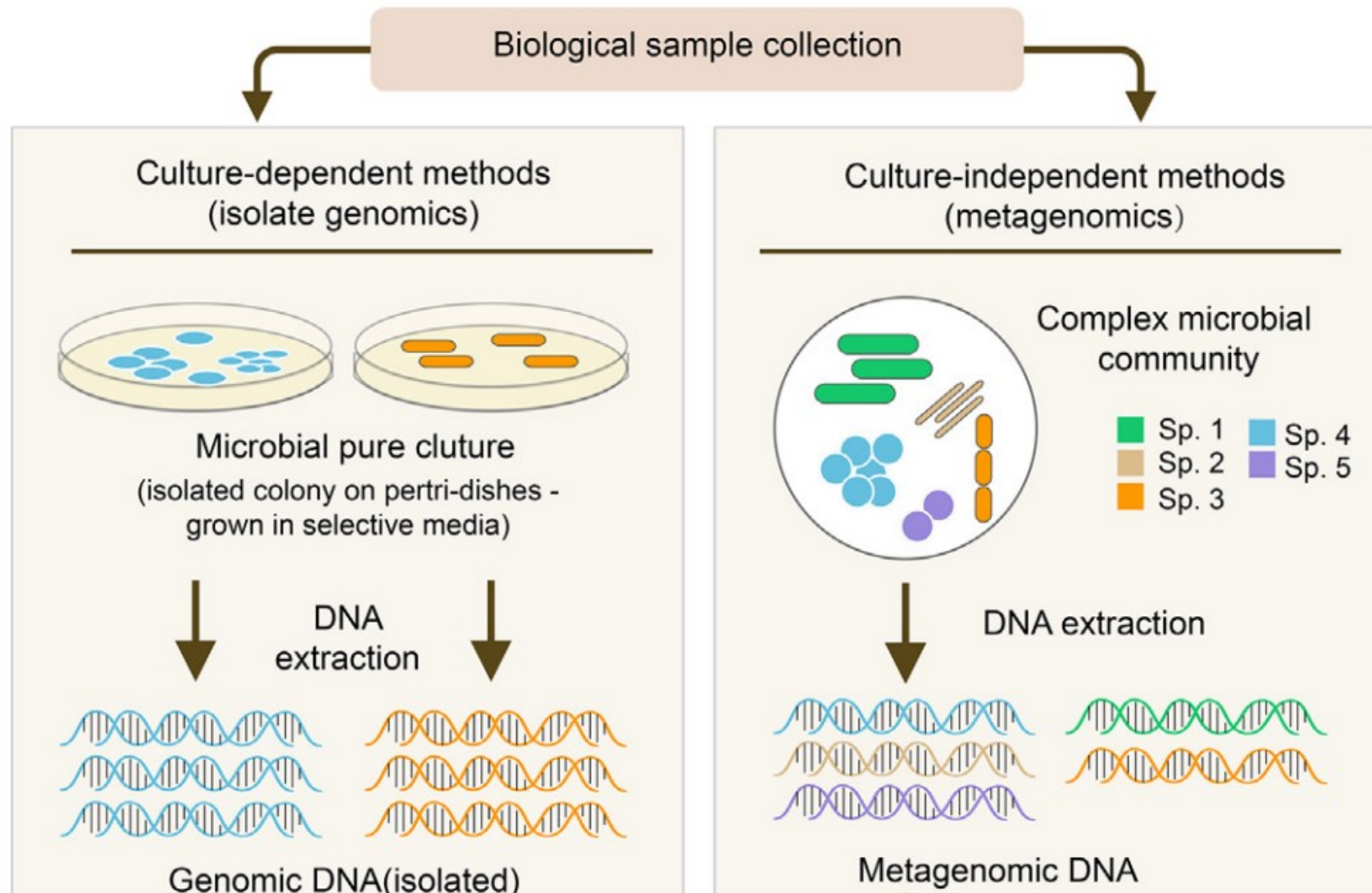
(b)

	Sp 1	Sp 2	Sp 3	Sp 4
Sp 1	0	0.5	0.3	0.6
Sp 2	0.5	0	0.5	0.5
Sp 3	0.3	0.5	0	0.4
Sp 4	0.6	0.5	0.4	0

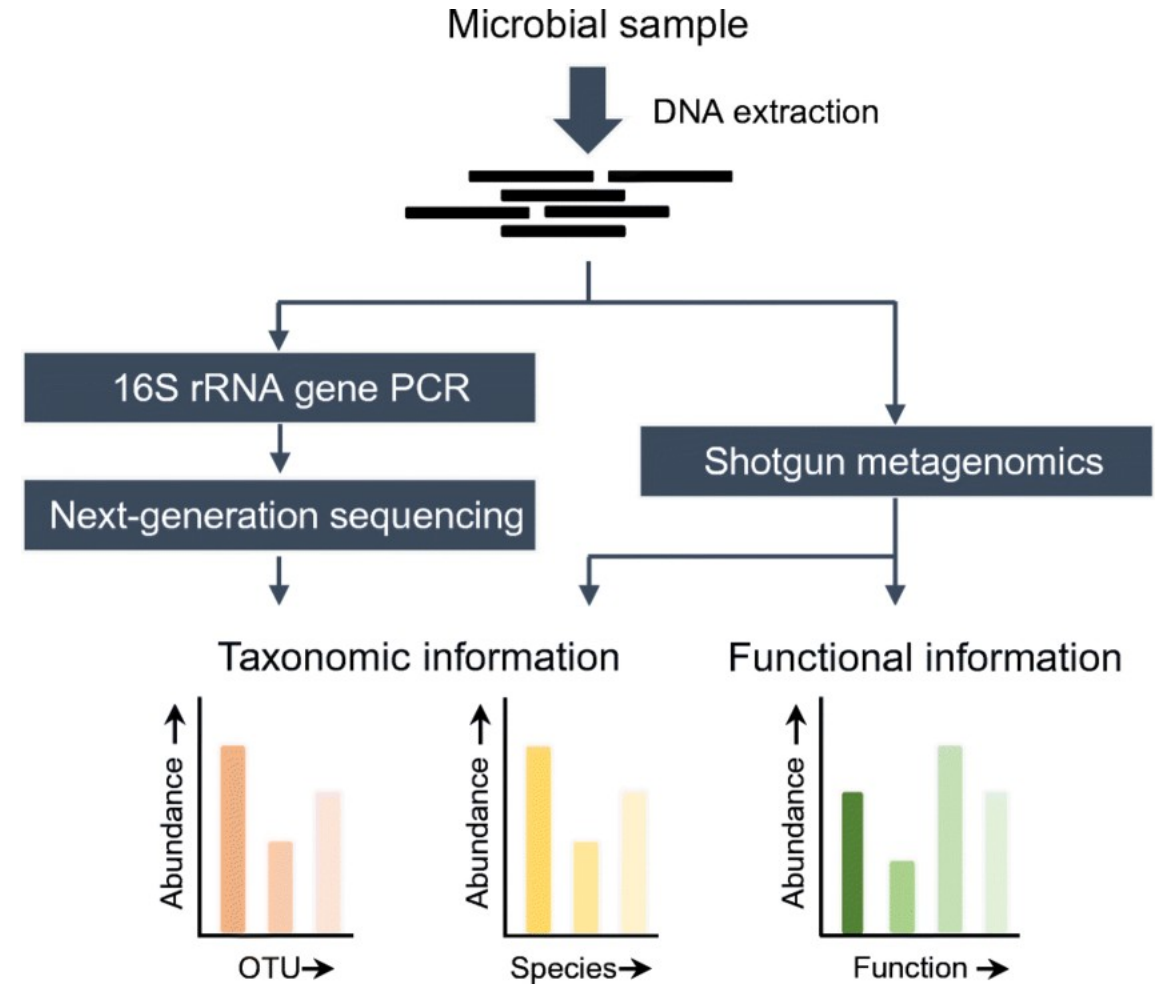
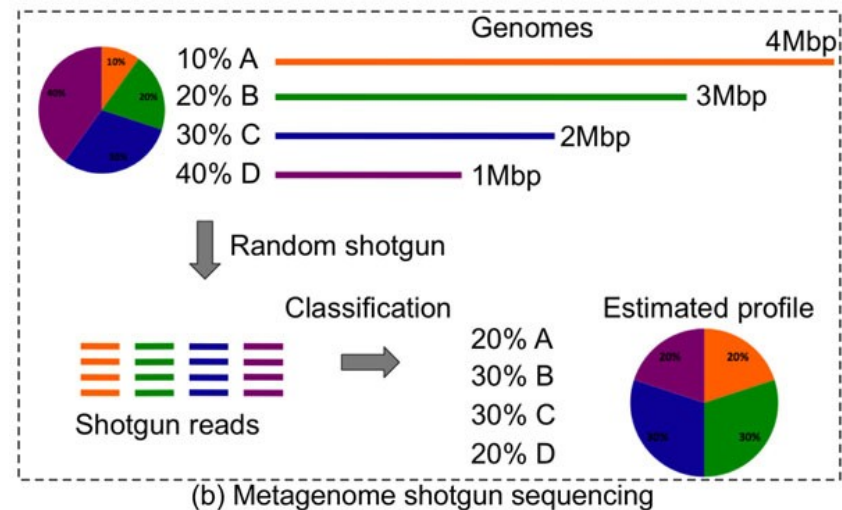
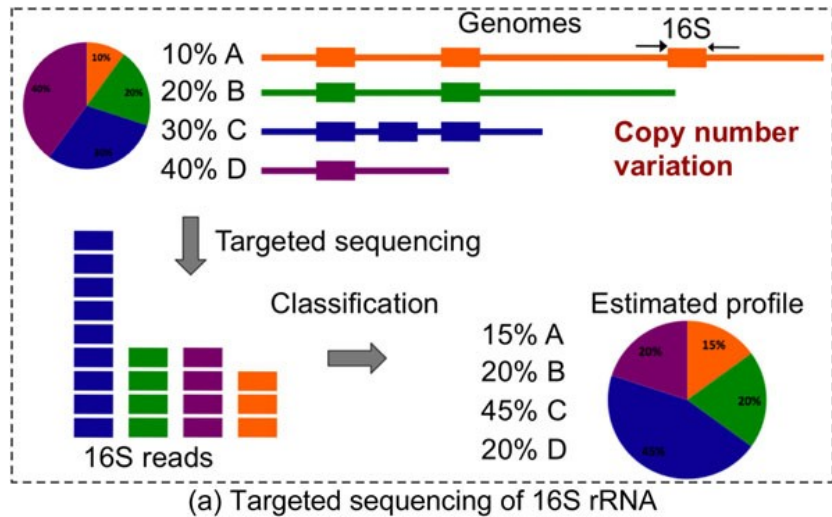
Building Phylogenetic Tree



Phân tích một bộ gen so với nhiều bộ gen



Target vs Shotgun Microbiome Analysis



Precision microbiome testing: STI, HPV and AMR

Women's Health Test

Page 1 of 4

Patient Name:	nan	Provider:	Mony Sary, Yap Chew, Wendy Ullmer	Patient ID:	VS8
Gender:	nan	Provider NPI:	nan	Specimen ID:	KH20.45474
DOB:	nan	Order Date:	nan	Specimen Type:	nan
Age:	nan	Health Status Reported:	nan	Collection Date:	nan

Sexually Transmitted Infections

Name	Associated Condition	Result
<i>Neisseria gonorrhoeae</i>	Gonorrhea, urethritis, pelvic inflammatory disease, gonococemia, gonococcal ophthalmia neonatorum	Detected
<i>Chlamydia trachomatis</i>	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
<i>Mycoplasma genitalium</i>	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
<i>Treponema pallidum</i>	Syphilis	Not Detected
<i>Haemophilus ducreyi</i>	Chancroid	Not Detected
<i>Trichomonas vaginalis</i>	Trichomoniasis	Not Detected
<i>Human papillomavirus</i>	Cervical and anogenital cancers, genital warts	Detected
<i>Herpes simplex virus</i>	Genital herpes, oral herpes	Not Detected

Viruses Detected

Name	Associated Condition
<i>Human papillomavirus 62 (HPV 62)</i>	Unknown risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

Antimicrobial Resistance Genes Detected

AMR Gene Name	Function	Drug Class
<i>Neisseria.gonorrhoeae.folP</i>	Dihydropteroate synthase (mutated)	Sulfonamide

Women's Health Test

Page 1 of 4

Patient Name:	nan	Provider:	Yap Chew	Patient ID:	nan
Gender:	nan	Provider NPI:	nan	Specimen ID:	202122865
DOB:	nan	Order Date:	nan	Specimen Type:	nan
Age:	nan	Health Status Reported:	nan	Collection Date:	nan

Sexually Transmitted Infections

Name	Associated Condition	Result
<i>Neisseria gonorrhoeae</i>	Gonorrhea, urethritis, pelvic inflammatory disease, gonococemia, gonococcal ophthalmia neonatorum	Not Detected
<i>Chlamydia trachomatis</i>	Chlamydia, cervicitis, urethritis, pelvic inflammatory disease	Not Detected
<i>Mycoplasma genitalium</i>	Urethritis, cervicitis, pelvic inflammatory disease	Not Detected
<i>Treponema pallidum</i>	Syphilis	Not Detected
<i>Haemophilus ducreyi</i>	Chancroid	Not Detected
<i>Trichomonas vaginalis</i>	Trichomoniasis	Not Detected
<i>Human papillomavirus</i>	Cervical and anogenital cancers, genital warts	Detected
<i>Herpes simplex virus</i>	Genital herpes, oral herpes	Not Detected

Viruses Detected

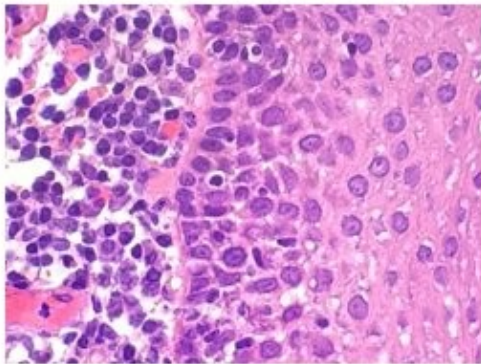
Name	Associated Condition
<i>Human papillomavirus 52 (HPV 52)</i>	High-risk for cervical cancer
<i>Human papillomavirus 68 (HPV 68)</i>	High-risk for cervical cancer

Note: Human papillomavirus (HPV) 16, 18, 31, 33, 35, 39, 45, 51, 56, 58, 59, and 68 are considered high-risk or probable high-risk due to their association with cervical cancer. HPV 6, 11, 42, 43, and 44 are considered low-risk for cervical cancer but may cause genital warts. Other HPV genotypes found in the sample may have intermediate or unknown risk for cervical cancer.

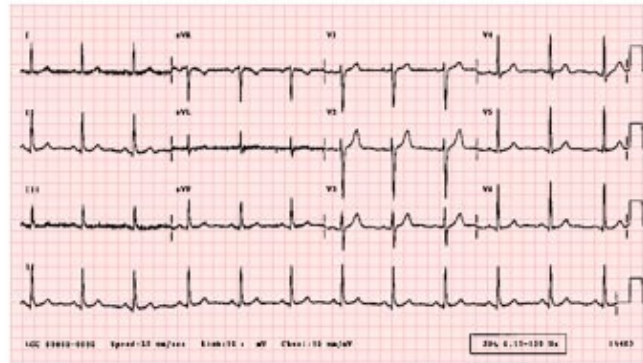
Ứng dụng AI trong phân tích dữ liệu hình ảnh y khoa

Dữ liệu ảnh giải phẫu bệnh và chẩn đoán hình ảnh

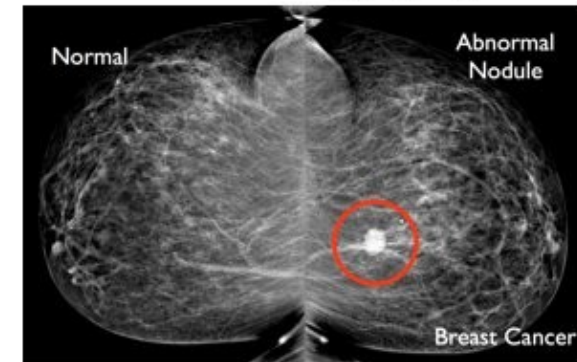
1. HE



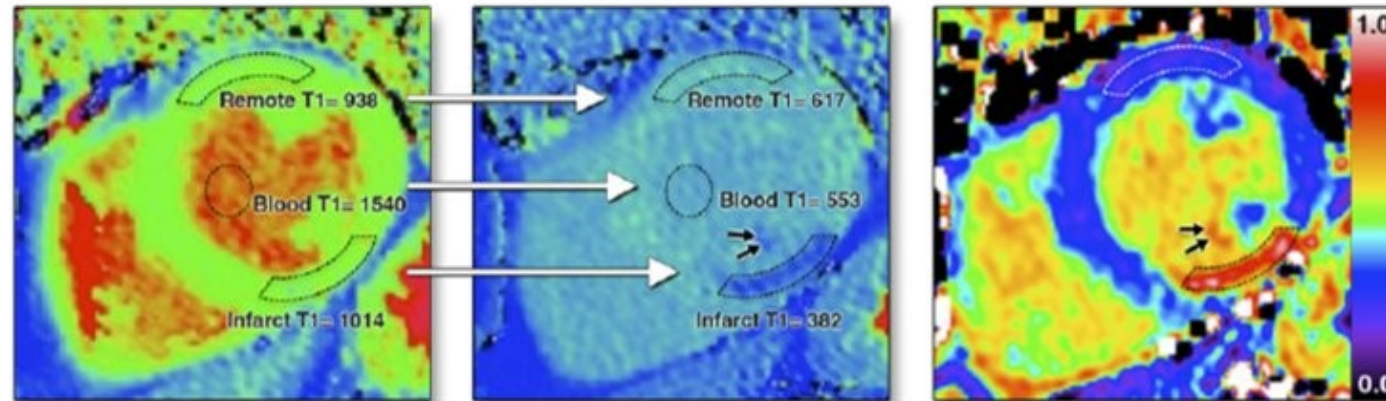
2. ECG



3. Mammography



4. Heart MRI



Người ngoài nghề nghĩ ...

CHỈ SỐ HUYẾT ÁP CHUẨN (Tiêu chuẩn của Tổ Chức y tế thế giới - WHO)

Độ tuổi	NAM		NỮ	
	Huyết áp tối đa	Huyết áp tối thiểu	Huyết áp tối đa	Huyết áp tối thiểu
15-19	120	70	111	67
20-29	124	75	114	69
30-39	126	79	118	73
40-49	130	83	126	78
50-59	137	85	134	81
60-69	143	84	139	81
70 trở lên	145	82	146	79

Microlife - «Hãy theo dõi huyết áp thường xuyên. Vì tăng huyết áp là kẻ giết người thầm lặng!»

Người ngoài nghề nghĩ ...

The diagram illustrates a data table with annotations. The word "Columns" is at the top with arrows pointing to the column headers: Name, Team, Number, Position, and Age. The word "Rows" is on the left with arrows pointing to the row indices: 0, 1, 2, 3, 4, 5, and 6. The word "Data" is at the bottom with a bracket pointing to the data cells of rows 2 through 6. The table itself has a light green background and contains the following data:

	<i>Name</i>	<i>Team</i>	<i>Number</i>	<i>Position</i>	<i>Age</i>
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Diagram illustrating a data table structure with annotations:

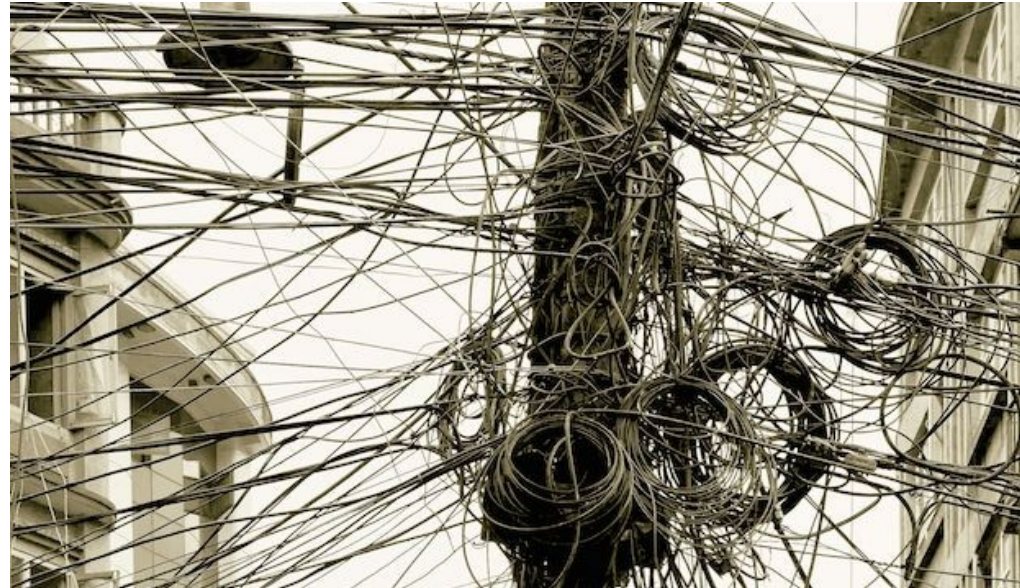
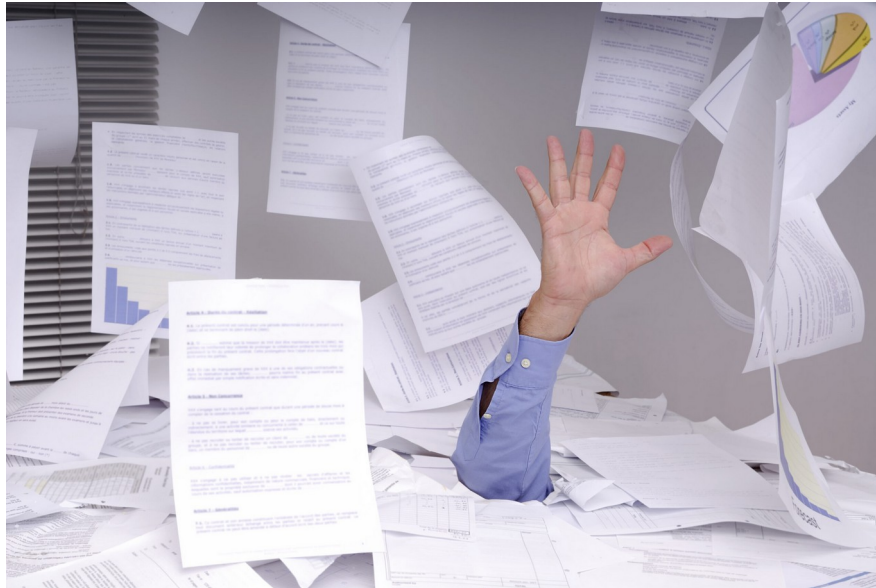
- Columns:** Name, Team, Number, Position, Age
- Rows:** 0, 1, 2, 3, 4, 5, 6
- Data:** The table contains data for 7 rows (0-6) and 5 columns.

Table Data:

Row	Name	Team	Number	Position	Age
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Logo: DG

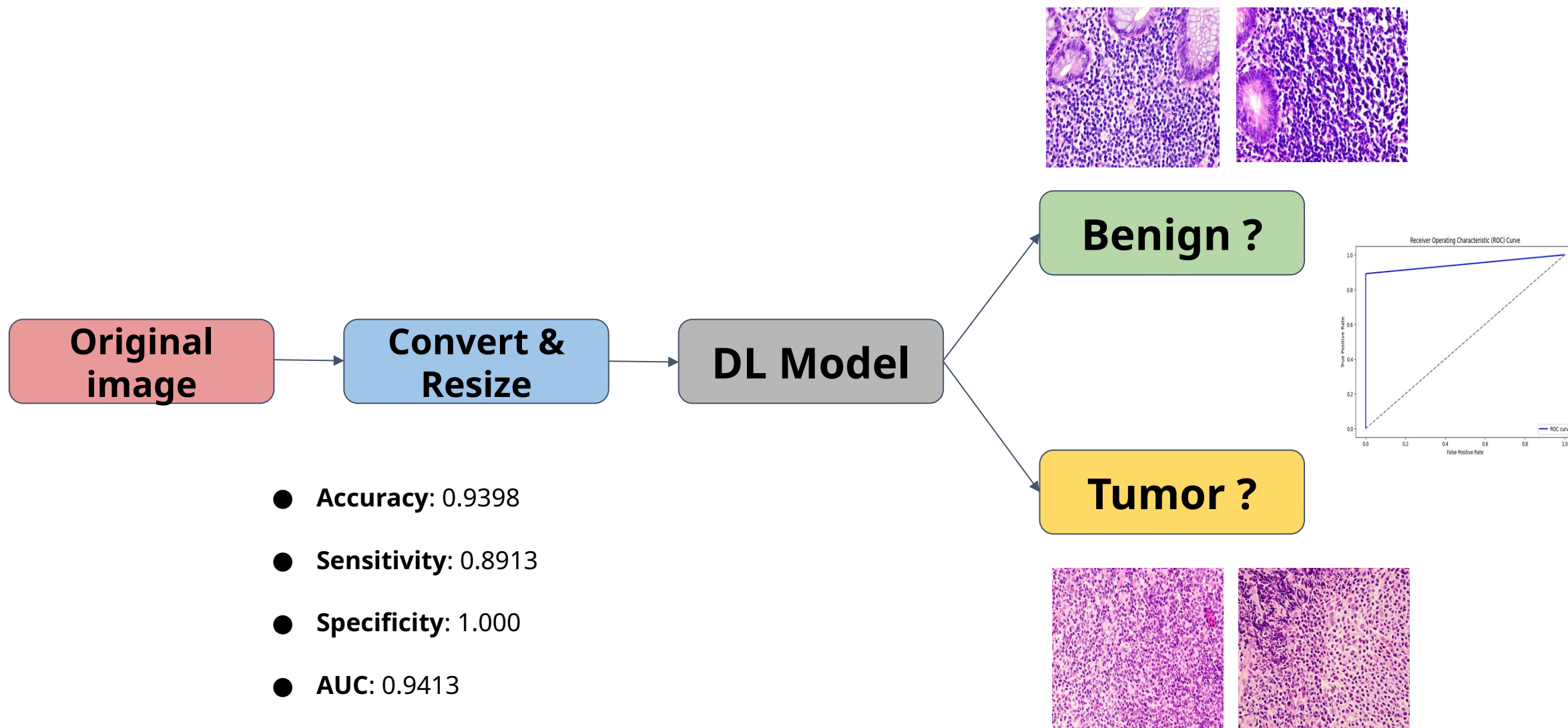
Thật sự khi mình thật sự bắt tay vào ...



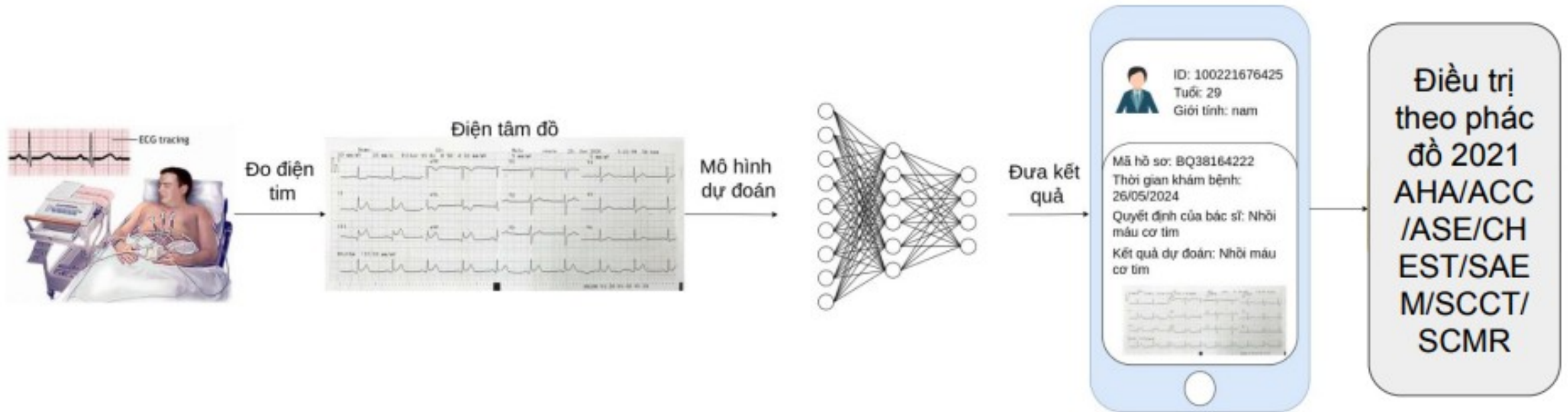
Thu thập, xây dựng, lưu trữ và tối ưu hóa các hệ thống dữ liệu (**Data engineer**)

- Kiến thức chuyên môn
- Kiến thức về cơ sở dữ liệu (SQL and Non-SQL)
- Kỹ năng lập trình
- Hadoop

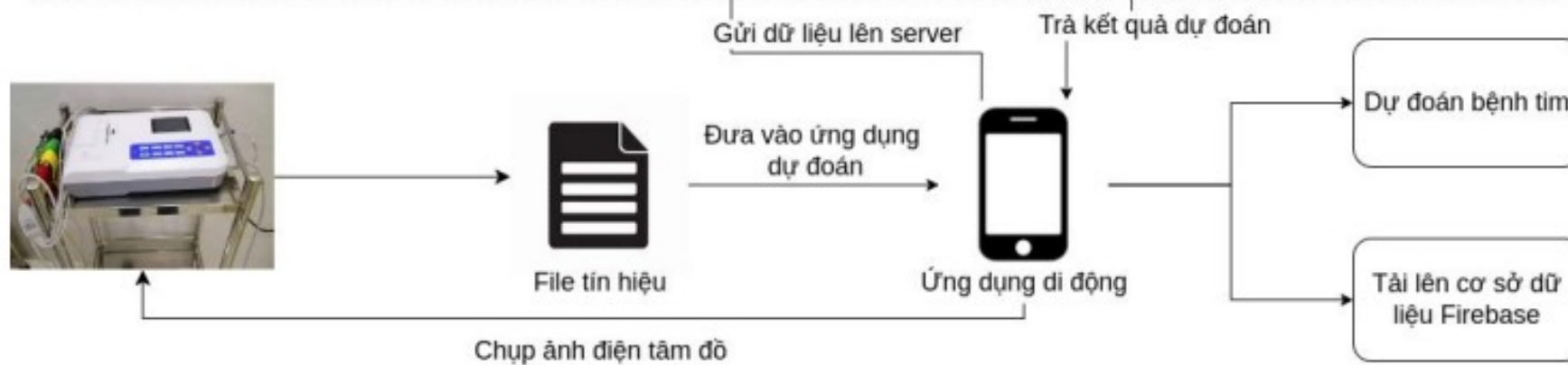
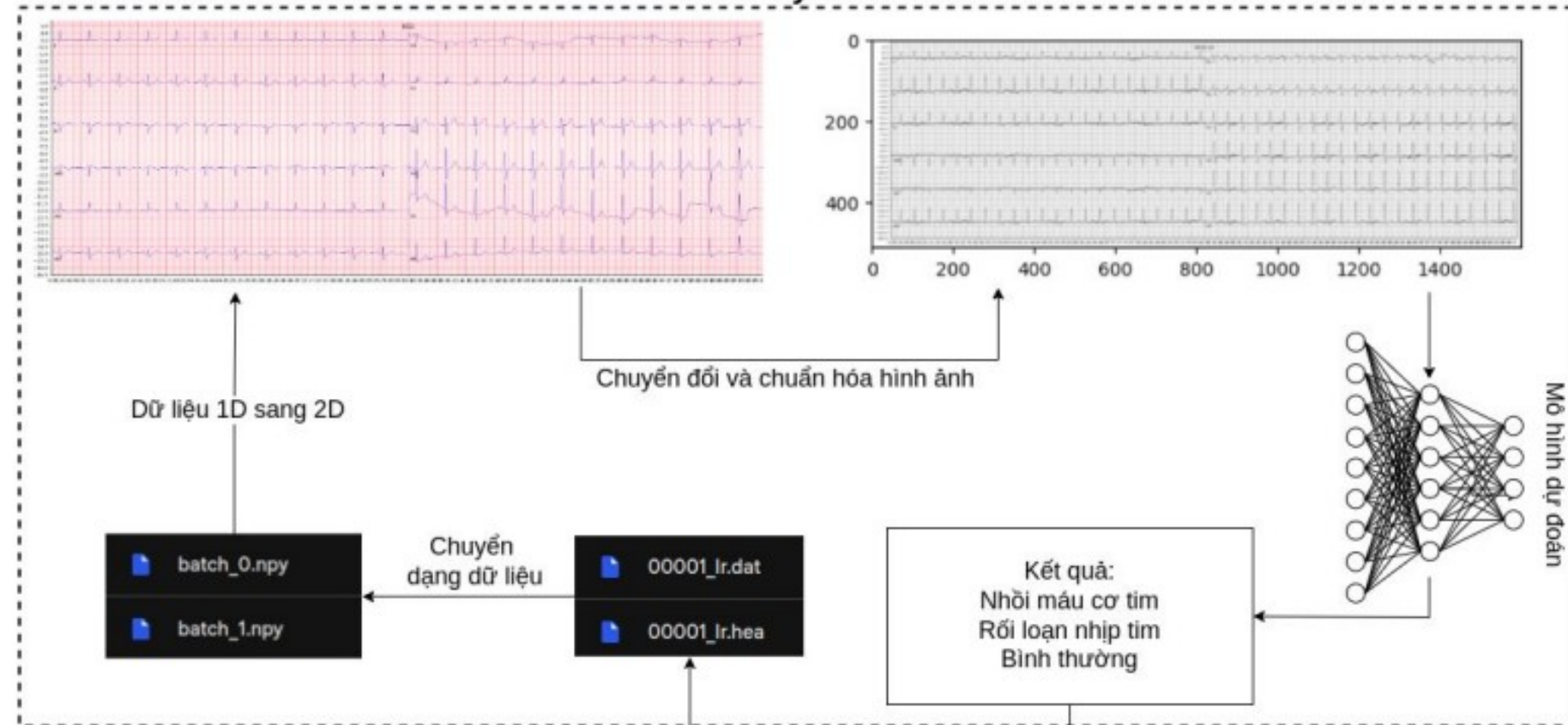
DeepHE: Lymphôm đường tiêu hóa



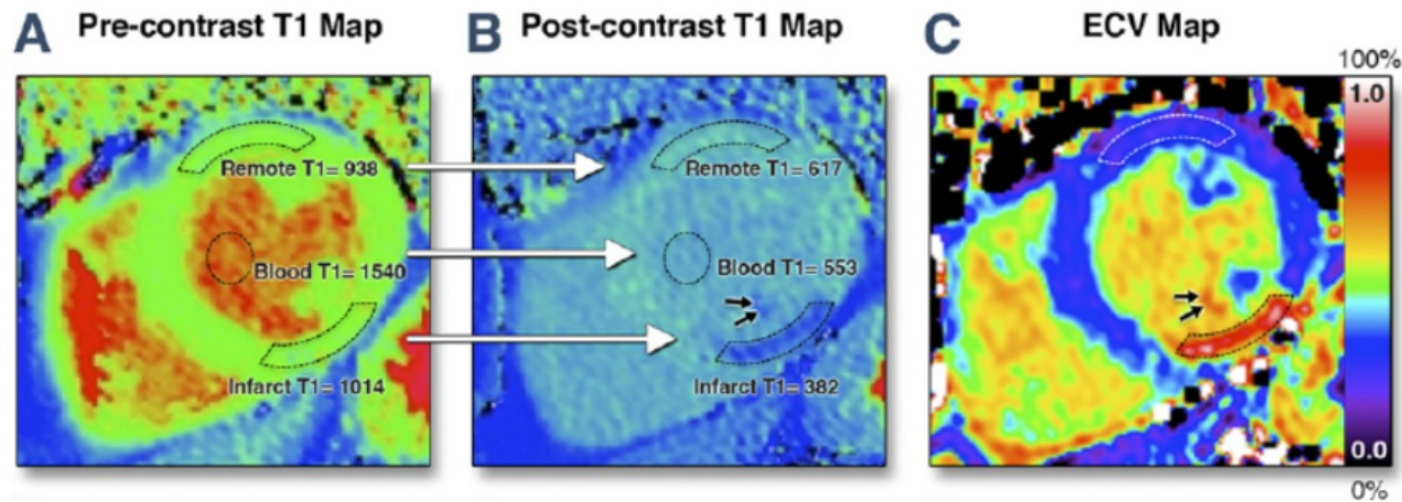
DeepECG: các bệnh về tim mạch



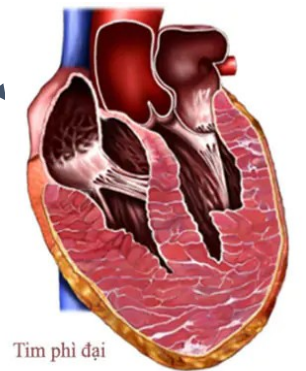
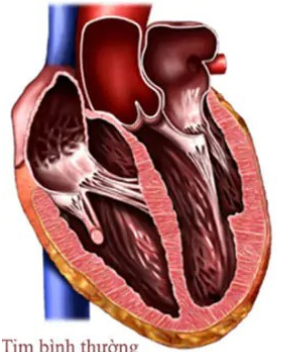
Máy chủ



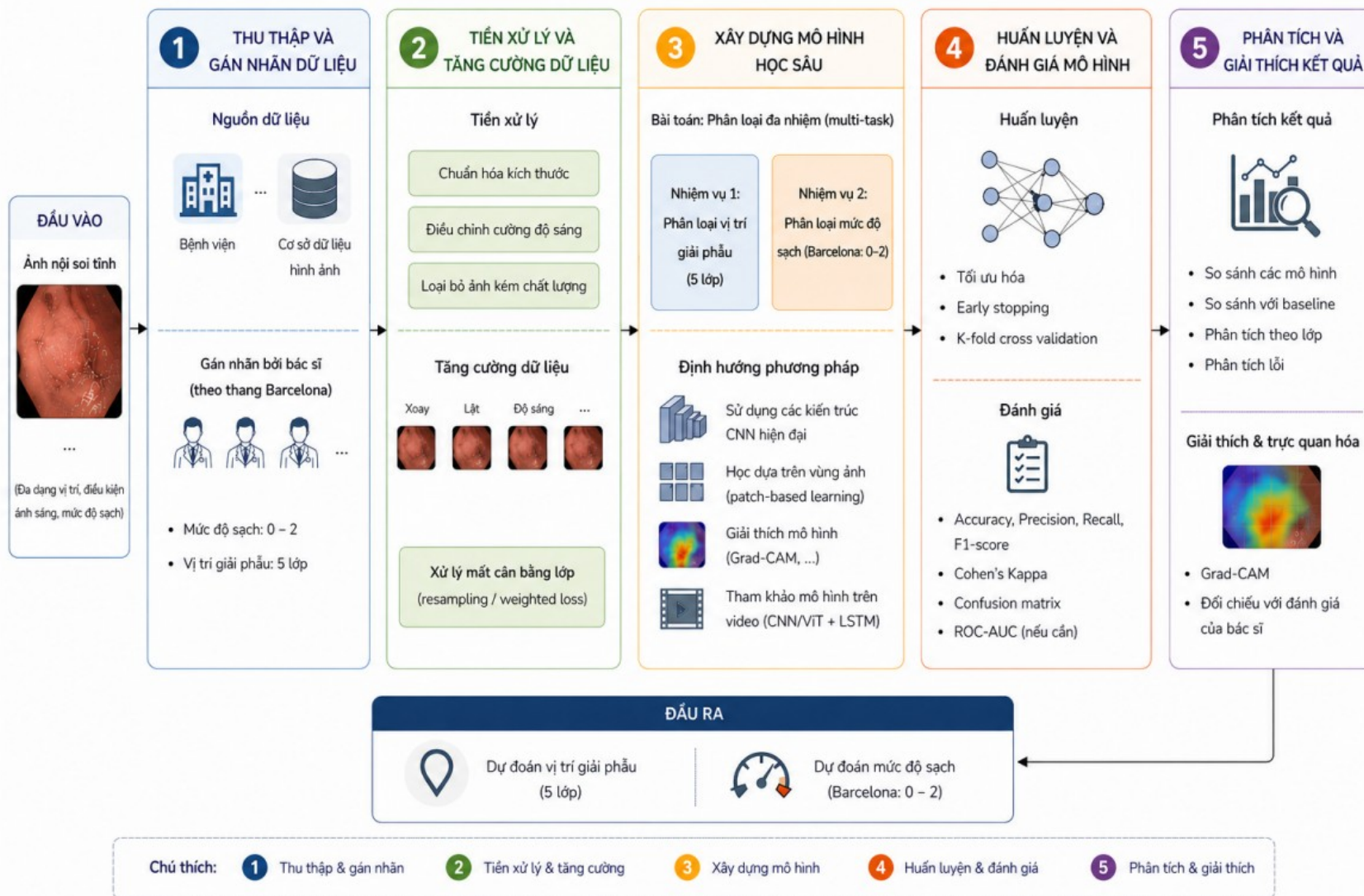
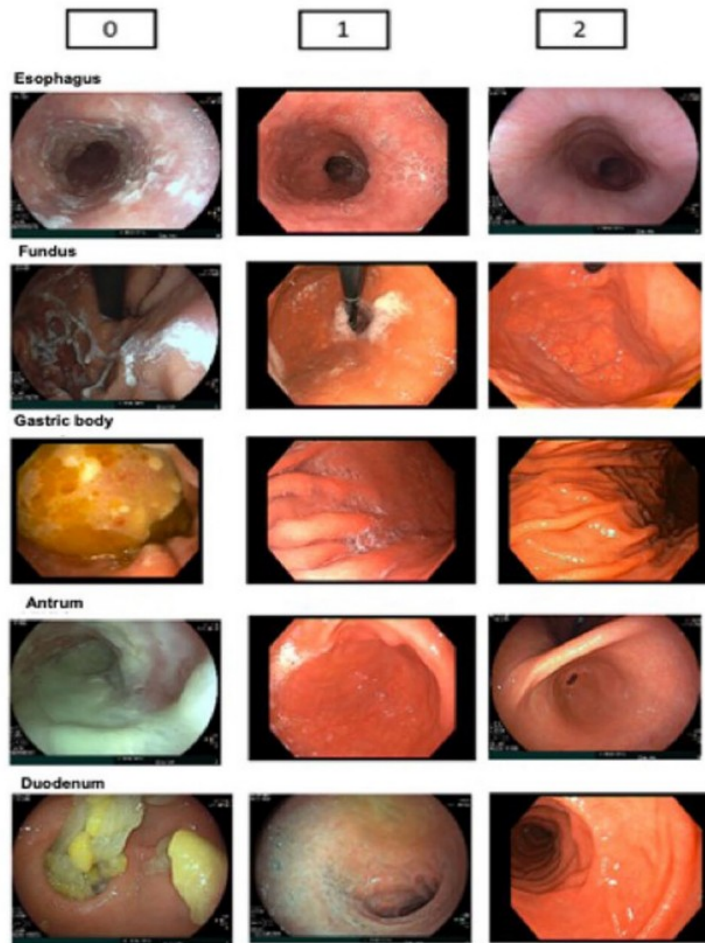
VECV: bệnh cơ tim phì đại



DL Model



Nội soi: ruột có sạch không?



Phosphate-to-albumin ratio and 30-day mortality in critically ill acute pancreatitis: multicentre development and external validation of an interpretable machine learning model

COHORTS & POPULATION MULTICENTRE COHORTS (N=2,226)

Critically Ill Acute Pancreatitis Patients
(N=2,226)

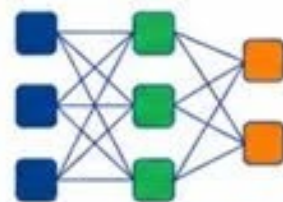
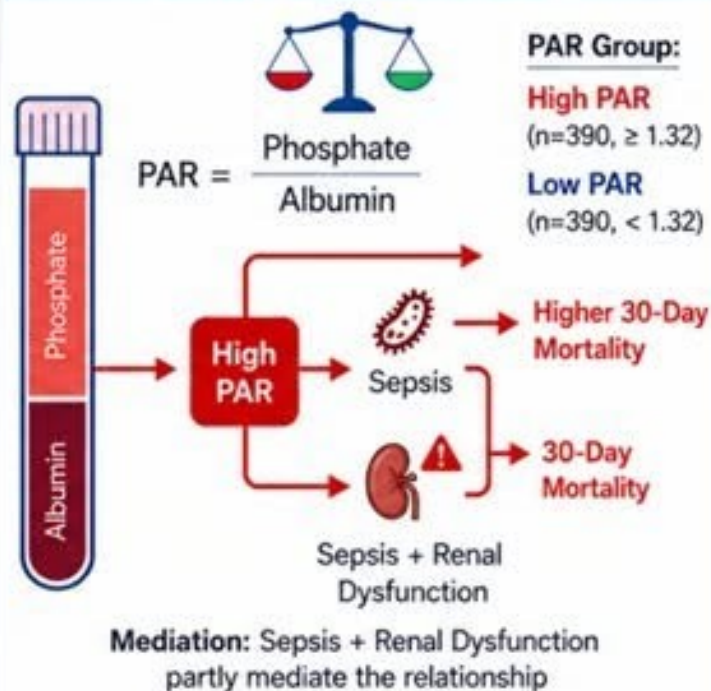
Training Cohort
(MIMIC-IV, n=780)

Validation Cohorts:
(n=609 [MIMIC-III],
n=837 [eICU])



Primary Outcome:
30-Day All-Cause
Mortality

EXPOSURE & MEDIATION ANALYSIS



Finalized and Deployed:
based on risk model
User-Friendly
Bedside Tool

CLINICAL KEY FINDINGS & MODEL DEPLOYMENT

Low PAR
High PAR

High PAR: Independent
Predictor of 30-Day
All-Cause Mortality
(aHR 1.94, 95% CI 1.11–3.38)

Model Performance (AUC)



**PAR-Integrated
LightGBM Model**
Good Discrimination
& Calibration

Clinical Index

Age
Blood pressure (mmHg)
Temperature (°C)
Respiratory rate

PAR

Age
Labs
Comorb.

RISK: 25%

Biochemical Index

Anion Gap (meq/L)
Phosphate (mg/dL)
Albumin (g/dL)
Lactate (mmol/L)
Prothrombin Time (sec)
Red Cell Distribution (%)

PAR-Index Web Calculator:
Bedside Prediction for Routine Practice



**Xin chân thành cảm ơn
sự lắng nghe của quý phụ
huynh và các tân sinh viên!**

Luu Phuc Loi, PhD

luu.p.loi@gmail.com

Phone/Zalo: 0901802182